

Contact

d.melconian@outlook.com

www.linkedin.com/in/

dariomelconian (LinkedIn)

medium.com/@dario61melconian
(Blog)

Top Skills

Transfer Learning

Large Language Models (LLM)

Natural Language Processing (NLP)

Languages

Italian

Spanish (Limited Working)

Certifications

Develop Generative AI Applications

2nd Place - DataQuest AI
Competition

Build RAG Applications

Applied Natural Language
Processing

Process Mining Course

Dario Melconian

Data Scientist at BMO Private Wealth

Toronto, Ontario, Canada

Summary

Data Scientist and AI practitioner with 4+ years of experience building production-grade machine learning and NLP systems at the intersection of technical depth and business impact. At BMO, operating as both hands-on data scientist and internal data science consultant, translating complex analytical problems into scalable solutions for executive stakeholders across Wealth Management. Work spans the full model lifecycle: from feature engineering and ML pipeline development to deployment and stakeholder communication. Recent highlights include an NLP summarization engine, a cross-sell recommendation system, and a multi-classification attrition model, driving measurable revenue impact across multiple lines of business. Beyond the day job, actively building in the LLM and agentic AI space across the full stack of modern agent frameworks. Projects span multi-model orchestration systems with iterative refinement loops (OpenAI, Claude, Ollama, Groq), autonomous research agents built on LangChain and Pydantic, multi-agent crews using CrewAI for financial analysis and portfolio reasoning, stateful conversational agents with memory and tool use via LangGraph, and distributed multi-agent systems with AutoGen. Most recently, building with the Model Context Protocol (MCP) to connect LLMs to external data sources and APIs in production-ready architectures. Work is grounded in real-world financial domains such as equity analysis, wealth management, and compliance, reflecting a focus on agentic systems that are not just technically sound but commercially deployable.

Experience

BMO Private Wealth

4 years

Data Scientist

April 2023 - Present (3 years 1 month)

Junior Data Scientist
May 2022 - April 2023 (1 year)

FGF Brands

Data Scientist

September 2021 - April 2022 (8 months)

Toronto, Ontario, Canada

- Trained machine learning algorithm for Human Detection on Custom Vision, tagging images inside the company's main plant, achieving a 96% accuracy and 89% performance score, exporting model using TensorFlow.
- Co-designed an algorithm for a Piece Counting project that used statistical analysis to detect and send events to SQL server if the number of food items traveling through different inspection points was over/under the appropriate statistically calculated range of count for any given camera at any minute interval, at live time.
- Set up and programmed a NVIDIA single board computer (SBC) from scratch, running OpenCV algorithms through Linux, with hardware connections such as a 4K camera capturing human face detection at live-time.

Actuarial and Statistical Undergraduate Association at Western University

Board Member / Mentor

October 2019 - August 2021 (1 year 11 months)

Education

Western University

Bachelor of Science Hon. Spec. - BS, Data Science · (2018 - April 2023)